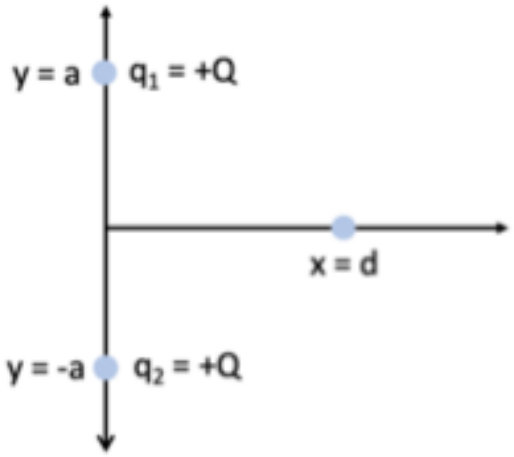
SPEC Problem Number 1:

Two charges, $q_1 = +Q$ and $q_2 = +Q$, are each placed a distance $y = a$ meters above and below the origin (thus on the y-axis). We will examine a point P on the x-axis at a distance of d from the origin. Refer to the diagram below:



A) In terms of the given quantities and fundamental constants, find the net electric field at point P. Be sure to explain any applied symmetry arguments.

B) Consider if P is very far away from the origin, at a distance d >> a (which means "d" is significantly larger than "a"). Provide an equation that approximates the net electric field at this point. How does this differ from your answer in part A) Why? Explain

Now suppose $q_1 = -Q$.

C) How do your answers in part A) change?

D) How does your answer in part B) change? This is known as the electric field due to a dipole. Simplify your expression using p = qD, where "p" is known as the electric dipole moment, "q" is charge, and "D" is the distance between the charges.